

ENGINEERING

ServicePro Delivery Roadmap

1. Current State

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Document Control

Purpose	Enterprise platform overview and buyer evaluation reference
Audience	Business leaders, platform administrators, evaluators, partners, and technical stakeholders
Scope	Capabilities, architecture, security, deployment, adoption, outcomes, and terminology
Source	ServicePro repository documentation; technical meaning preserved

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Version: 8.0.0-alpha.1 Date: July 2026 Planning Horizon: 6 months

ServicePro is a functional alpha with strong feature coverage for field service operations. The platform serves real tenants (Aqua Pro Plumbing, C & D Landscaping) but runs on free-tier infrastructure without production hardening. The gap between "working code" and "production-grade platform" is the focus of this roadmap.

2. Workstream Overview

#	Workstream	Priority	Timeline	Objective
1	Production Hardening	Critical	Weeks 1–4	Eliminate service reliability risks
2	Developer Experience	High	Weeks 2–6	CI/CD, testing, type safety
3	Frontend Modernization	High	Weeks 4–10	Performance, maintainability, UX
4	API Architecture	Medium	Weeks 6–14	Framework adoption, router refactor
5	Data Platform	Medium	Weeks 8–16	Backups, monitoring, connection pooling
6	Observability	Medium	Weeks 10–18	Error tracking, metrics, alerting
7	Security Hardening	High	Weeks 4–12	Scanning, rotation, penetration testing
8	Feature Completion	Ongoing	Weeks 1–24	Customer portal, mobile, admin UI

3. Workstream Details

3.1 Production Hardening (Weeks 1–4)

Goal: Make the platform reliable enough for paying customers.

Task	Effort	Impact
Upgrade Render API to Starter plan (\$7/mo)	1 hour	Eliminates 50s cold starts
Upgrade Supabase to Pro (\$25/mo)	1 hour	Daily backups, no pause, 8 GB
Add health check alerting (UptimeRobot or similar)	2 hours	Know when service is down
Configure proper CORS for production domain only	1 hour	Tighten security surface
Add package-lock.json to root and enforce npm ci	1 hour	Reproducible builds
Move JWT_SECRET to environment variable (not generate)	1 hour	Survives redeploy
Document recovery procedure for DB restoration	4 hours	Runbook for incidents

Exit Criteria: API responds in <2s on first request, database has daily backups, alerting notifies on downtime.

3.2 Developer Experience (Weeks 2–6)

Goal: Prevent regressions, enforce quality, enable safe collaboration.

Task	Effort	Impact
Add TypeScript --noEmit check to CI	2 hours	Catch type errors before deploy
Run test suite in CI (even if some fail, report)	4 hours	Visibility into test health

Task	Effort	Impact
Add ESLint with recommended rules	4 hours	Code consistency
Enable branch protection on main	30 min	Require PR for changes
Add PR template with checklist	1 hour	Structured review
Set up Playwright for E2E smoke tests	8 hours	Critical path verification
Create .env.local.example for onboarding	1 hour	New developer setup
Document local development setup in README	2 hours	Reduce onboarding friction

Exit Criteria: CI blocks PRs with type errors, at least 5 E2E tests covering login/jobs/storefront, branch protection enabled.

3.3 Frontend Modernization (Weeks 4–10)

Goal: Improve performance, reduce component complexity, establish design system.

Task	Effort	Impact
Break StorefrontBuilder into 5–8 sub-components	12 hours	Maintainability
Break PublicStorefront into section components	8 hours	Testability
Extract shared UI primitives (Button, Input, Modal, Card)	16 hours	Consistency
Remove CSS !important overrides — use proper specificity	8 hours	Predictable styling
Add component-level Storybook or similar catalog	8 hours	Visual regression
Implement proper loading/error states for all data fetches	8 hours	UX polish
Add skeleton screens for slow API responses	4 hours	Perceived performance
Audit accessibility (WCAG 2.1 AA) for core flows	8 hours	Compliance

Exit Criteria: No component exceeds 400 lines, shared UI library covers 80% of patterns, no !important in stylesheets.

3.4 API Architecture (Weeks 6–14)

Goal: Replace fragile raw HTTP routing with maintainable framework.

Task	Effort	Impact
Evaluate Express vs. Fastify (recommend Fastify for perf)	4 hours	Informed decision
Extract route groups from router.js into modular files	16 hours	Maintainability
Adopt chosen framework incrementally (adapter pattern)	24 hours	Modern middleware
Remove or lazy-load Phase 9–45 stub modules	8 hours	Faster startup
Standardize error response format across all routes	8 hours	API consistency
Generate OpenAPI spec from route definitions	8 hours	Documentation
Add request/response validation (Zod or Joi)	16 hours	Input safety

Exit Criteria: Router.js under 200 lines, all routes registered via framework router, startup time under 3 seconds.

3.5 Data Platform (Weeks 8–16)

Goal: Production-grade database operations.

Task	Effort	Impact
Implement connection pooling (Supabase Vibes or PgBouncer)	4 hours	Connection efficiency
Add slow query logging (pg_stat_statements)	4 hours	Performance visibility
Set up automated backup verification (restore to temp)	8 hours	Backup confidence
Add database migration CI check (dry-run)	4 hours	Prevent broken migrations
Implement proper migration versioning (sequential numbered)	8 hours	Schema history
Add tenant data isolation tests	8 hours	Security verification
Plan upgrade path from free tier when needed	2 hours	Capacity planning

Exit Criteria: Connection pooling active, slow queries logged, backup restore verified monthly.

3.6 Observability (Weeks 10–18)

Goal: See what's happening in production without SSH access.

Task	Effort	Impact
Add Sentry for error tracking (free tier: 5K events/mo)	4 hours	Error visibility
Add structured JSON logging	8 hours	Log searchability
Implement request duration tracking	4 hours	Performance baseline
Add database query timing	4 hours	Bottleneck detection
Set up Render metrics dashboard	2 hours	Resource monitoring
Add custom health metrics (tenant count, job volume)	4 hours	Business metrics
Configure alerting thresholds	4 hours	Proactive notification

Exit Criteria: All unhandled errors captured, P95 latency visible, alerts for error spike and downtime.

3.7 Security Hardening (Weeks 4–12)

Goal: Close security gaps before acquiring more customers.

Task	Effort	Impact
Add npm audit to CI with failure on high severity	2 hours	Dependency safety
Implement JWT secret rotation mechanism	8 hours	Secret hygiene
Add CSRF protection for state-changing endpoints	8 hours	Request forgery prevention
Implement rate limiting per tenant (not just global)	4 hours	Abuse prevention
Add input sanitization audit for all user inputs	8 hours	Injection prevention
Enable Supabase Row Level Security for shared tables	12 hours	Defense in depth
Schedule quarterly dependency update review	2 hours	Maintenance cadence
Document security incident response procedure	4 hours	Readiness

Exit Criteria: Zero high-severity npm audit findings, per-tenant rate limiting active, RLS on shared tables.

3.8 Feature Completion (Ongoing)

Goal: Complete customer-facing features that drive revenue.

Feature	Effort	Priority
Customer Portal — full self-service (booking, payments, history)	40 hours	High
Admin UI — complete platform management interface	24 hours	High
Mobile responsive improvements (remaining pages)	16 hours	Medium
Email notifications (transactional via SendGrid/Resend)	12 hours	High
Payment processing integration (Stripe)	20 hours	High
SMS notifications (Twilio)	8 hours	Medium
Reporting export to PDF/Excel	8 hours	Medium
Multi-user support per workspace	12 hours	Medium

4. Resource Requirements

4.1 Infrastructure Costs (Monthly)

Service	Current	Recommended	Cost
Render API	Free	Starter	\$7/mo
Render DB	Free	— (use Supabase)	\$0
Supabase	Free	Pro	\$25/mo
Cloudflare Workers	Free	Free	\$0
Sentry	—	Free tier	\$0
UptimeRobot	—	Free tier	\$0
Total	\$0	—	\$32/mo

4.2 Scaling Thresholds

Trigger	Action	Cost Impact
>5 tenants active	Upgrade Render to Standard (\$25/mo)	+\$18/mo
>50 concurrent users	Add Redis caching	+\$15/mo
>1 GB database	Supabase already Pro	Included
>100 tenants	Evaluate multi-instance API	+\$50/mo

5. Milestone Schedule

Milestone	Target	Key Deliverables
M1: Stable Production	Week 4	Paid hosting, backups, alerting, no cold starts
M2: CI/CD Complete	Week 6	Tests in CI, branch protection, type checking
M3: Security Baseline	Week 12	Audit clean, RLS, rate limiting, rotation
M4: Framework Migration	Week 14	Fastify/Express adopted, router refactored

Milestone	Target	Key Deliverables
M5: Frontend V2	Week 10	Component library, no overrides, accessible
M6: Observability	Week 18	Full error tracking, metrics, alerting
M7: Feature Complete	Week 24	Customer portal, payments, email, mobile

6. Risk Register

Risk	Likelihood	Impact	Mitigation
Free tier service outage during customer demo	High	High	Upgrade to paid plans (M1)
Database pause loses customer data	Medium	Critical	Supabase Pro + backup verification
Regression deployed to production	High	Medium	CI test gating (M2)
Security breach via unvalidated input	Medium	Critical	Input validation audit (M3)
Developer confusion from 14K file count	High	Low	Documentation, .gitignore cleanup
Router.js becomes unmaintainable	High	Medium	Framework migration (M4)
Customer-visible errors with no alerting	High	Medium	Sentry integration (M6)

7. Decision Log

Decision	Rationale	Date
Stay on Supabase (not Render DB)	Better free tier, familiar, pooling built-in	July 2026
Static export for frontend	Cloudflare Workers = fast, free, global CDN	July 2026
No Express/Fastify yet	Working system, refactor after stabilization	July 2026
Phase 9–45 remain scaffolded	Future roadmap, don't delete, but lazy-load	July 2026
Monorepo stays (no microservices yet)	Single deploy target, simple, 2 tenants	July 2026